

Class 12: Pt2 (Population Analysis)

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Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```
data <- read.table("https://bioboot.github.io/bgg213_W19/class-material/rs8067378_ENSG000000000000.txt",
                  header = TRUE)
table(data$geno)
```

```
A/A A/G G/G
108 233 121
```

```
aggregate(exp ~ geno, data = data, median)
```

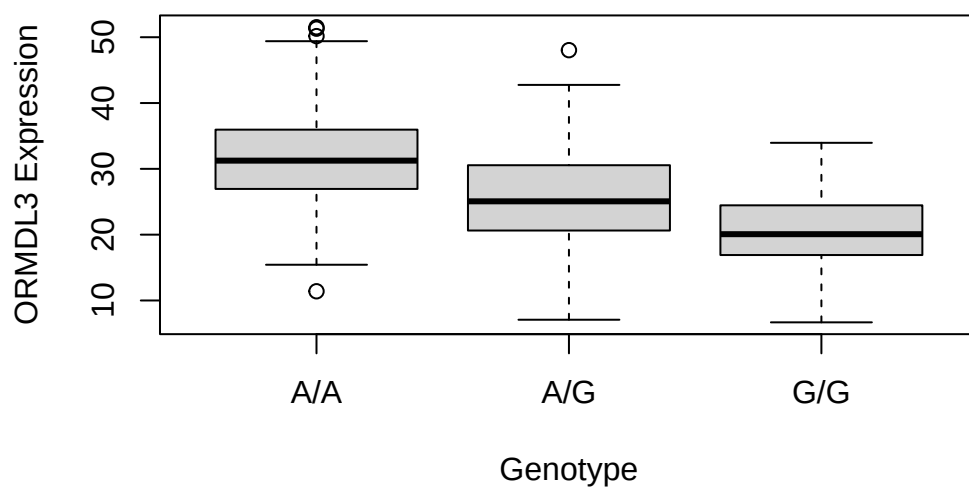
```
      geno      exp
1  A/A 31.24847
2  A/G 25.06486
3  G/G 20.07363
```

The rs8067378 ORMDL3 expressing file sample sizes were 108 A/A, 233 A/G, and 121 G/G. The median expression values were 31.24847 for A/A, 25.06486 for A/G, and 20.07363 for G/G.

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
boxplot(exp ~ geno,
        data = data,
        xlab = "Genotype",
        ylab = "ORMDL3 Expression",
        main = "ORMDL3 Expression by rs8067378 Genotype")
```

ORMDL3 Expression by rs8067378 Genotype



Yes, rs8067378 appears to affect ORMDL3 expression because individuals with the G/G genotype has lower ORMDL3 expression than individuals with A/A genotype.